# SILVIA ROSSI

**Senior Scientist**

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**CWI - Amsterdam, The Netherlands — Italian (native); English (fluent)**

Senior scientist with a PhD in Electronic and Electrical Engineering (UCL) and over seven years of experience in machine learning, predictive modelling and behavioural analytics. I study how people navigate, experience and interact in immersive environments: I build end-to-end Python pipelines for multimodal behavioural data and develop evaluation methods for interactive systems, including metric design and validation against reference results.

## TECHNICAL SKILLS

- **Machine learning and evaluation:** predictive modelling, clustering, classification/regression, deep learning (PyTorch, TensorFlow), statistical analysis, experimental design and research evaluation (evals).
- **Software engineering:** Python (pandas, NumPy, SciPy), MATLAB, HTML, C/C++ (basic); Git and GitHub, unit testing, deployment on Vercel.
- **AI-assisted workflows:** Claude Code for software development and research workflows; TypeScript, React and Vite in personal projects (below).

## PROFESSIONAL EXPERIENCE

**Senior Scientist** May 2024 - Present
*CWI, Distributed and Interactive Systems group, Amsterdam*

- Built end-to-end Python pipelines for multimodal behavioural data (movement, gaze, voice and decision timing) from multi-user virtual reality (VR) studies and group-behaviour datasets.
- Ported legacy MATLAB analysis code to Python; validated outputs against the original implementation using parity checks and sampling-rate sensitivity tests.
- Developed behavioural metrics for user modelling and system optimisation, combining statistical analysis with deep learning methods.
- Lead CWI activities in INDUX-R, including the task on scene orchestration, resource scheduling and real-time synchronisation; coordinate international technology partners.
- Co-supervise a PhD student on HoloCalls, a volumetric telepresence system; supervise MSc/PhD students and technical staff.

**Postdoctoral Researcher (including ERCIM Fellow)** May 2022 - April 2024
*CWI, Distributed and Interactive Systems group, Amsterdam*

- Developed clustering and predictive modelling approaches to analyse user behaviour and identify similarities not apparent from individual data streams.
- Designed controlled user studies and mixed-method analytical frameworks; contributed to the real-time social VR platform VR2Gather.
- Coordinated EU Horizon project tasks across 8–10 international researchers.

**Earlier research internships:** CWI (Oct 2020 - Mar 2021), behavioural analysis in immersive systems; UCL (Nov 2016 - Apr 2017), optimisation of VR content streaming.

## SOFTWARE PROJECTS (PERSONAL, 2026)

**Puntatime**, a TV-show tracking progressive web app (TypeScript, React, Vite, IndexedDB), and **Spesa & Menù**, a React/Vite meal planner with unit-tested CSV import; both built with Claude Code and deployed on Vercel.

## SELECTED RESEARCH ACHIEVEMENTS

- **Behavioural similarity and clustering for VR users** (ACM TOMM, 2020). Developed a user-affinity metric; findings informed processing and system-design strategies achieving up to **70% savings in encoding and storage costs** while improving user experience. Contributed a public dataset. *2021 Nicholas Georganas ACM TOMM Best Paper Award.*
- **Evaluation methodology:** Task Lead on metrics in TRANSMIXR (2023 - 2025), identifying and validating methods for evaluating extended reality (XR) user experience. First author of *A Cookbook of User Metrics in Extended Reality*, a framework for selecting behavioural, physiological and self-reported metrics (ACM Computing Surveys, under revision).
- **Other recognition:** QoMEX Best Paper Award (2025); ACM SIGMM Best PhD Thesis Award (2023); UCL Fabrizio Lombardi Prize for the best PhD thesis in Electronic & Electrical Engineering (2022); ERCIM Alain Bensoussan Postdoctoral Fellowship (2022).

Full publication list: silvia024.github.io/publications.html.

## EDUCATION

**PhD, Electronic and Electrical Engineering** May 2017 - Feb 2022
*University College London (UCL)*
User behaviour modelling, machine learning and optimisation for immersive communication systems.

**MSc, Electronic and Telecommunication Engineering for Energy**
*University of Bologna, Cesena Campus* 2013 - 2016
Grade: 110/110.

**Professional development (Coursera, 2026):** DeepLearning.AI PyTorch: Fundamentals; Google AI Fundamentals.

## LEADERSHIP AND COMMUNITY

Steering Committee Chair, ACM Multimedia, and SIGMM Director of Conferences (both since November 2024); General Co-Chair, ACM IMX 2027; co-organiser of an ACM Europe Seasonal School on Social XR (2025 - 2026).